WEEK 5

PROGRAM 1

```
#include <iostream>

using namespace std;

class Addition

{

private:

int n;

int *ptr;

int sum;

public:

void getSize()

{

cout << "Enter number of elements: ";

cin >> n;

ptr = new int[n];

}

void getElements()

{

cout << "Enter " << n << " elements:\n";

for(int i = 0; i < n; i++)
```

```
{
cin >> ptr[i];
}
}
void calculateSum()
{
sum = 0;
for(int i = 0; i < n; i++)
{
sum = sum + ptr[i];
}
}
void displaySum()
{
cout << "Sum of elements = " << sum << endl;
}
~Addition()
{
delete[] ptr;
}
};
```

```
int main()
{
Addition obj;
obj.getSize();
obj.getElements();
obj.calculateSum();
obj.displaySum();
return 0;
}
```

PROGRAM 2

```
#include <iostream>
using namespace std;
class Node
{
public:
int data;
Node *next;
};
int main()
{
```

```
Node *head = NULL, *second = NULL, *third = NULL, *fourth = NULL;
head = new Node();
second = new Node();
third = new Node();
fourth = new Node();
head->data = 10;
head->next = second;
second->data = 20;
second->next = third;
third->data = 30;
third->next = fourth;
fourth->data = 40;
fourth->next = NULL;
Node *temp = head;
cout << "Single Linked List Elements are:\n";
while(temp != NULL)
{
cout << temp->data << " -> ";
temp = temp->next;
}
cout << "NULL";
```

```
return 0;
}
```

PROGRAM 3

```
#include <iostream>
using namespace std;
class Node
{
public:
int data;
Node *next;
};
int main()
{
int n, value;
Node *head = NULL, *newnode = NULL, *temp = NULL;
cout << "Enter number of nodes: ";
cin >> n;
for(int i = 1; i <= n; i++)
{
newnode = new Node();
cout << "Enter data for node " << i << ": ";
```

```
cin >> value;
newnode->data = value;
newnode->next = NULL;
if(head == NULL)
{
head = newnode;
temp = head;
}
else
{
temp->next = newnode;
temp = newnode;
}
}
temp = head;
cout << "Single Linked List Elements are:\n"
;
while(temp != NULL)
{
cout << temp->data << " -> ";
temp = temp->next;
```

```
}
cout << "NULL";
return 0;
}
```